LECTURE 10

LAST TIME

LECTURE 9: SUMMARY

Risk-neutral probabilities

FTAP

THIS TIME

LECTURE 10 OUTLINE

Warm-up on risk-neutral probabilities and the FTAP

Portfolio choice fundamentals

Asset prices from supply and demand

WARM-UP

WHERE IS p?

In binomial model, priced any security X as $X_0 = \dfrac{qX_u + (1-q)X_d}{1+r}$

where $q = \dfrac{1+r-d}{u-d}$

Why doesn't price depend on probability of up p?

INTERPRETATION OF q

Suppose u and d equally likely ($p = 1/2$) but $q < 1/2$

Which state do you think represents a boom and which a recession?

UTILITY MAXIMIZATION

U MAXIMIZATION

U is function of consumption with with $U' > 0$ and $U'' < 0$

Max $U(c_0) + \delta \mathbb{E}[U(c_1)]$

c_0 is consumption; c_1 risky consumption tomorrow

δ is rate of time preference

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Otherwise would buy/sell securities changing (c_0, c_1)

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$$\implies X_0 = \frac{\delta \mathbb{E}\big[U'(c_1)X_1\big]}{U'(c_0)} \qquad (\star)$$

Often written $X_0 = \mathbb{E}[M_1 X_1]$ where $M_1 := \dfrac{\delta U'(c_1)}{U'(c_0)} \equiv \dfrac{\delta U_1'}{U_0'}$

M called "stochastic discount factor"

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Called "Euler equation" written as $U'(c_0) = (1+r)\delta\mathbb{E}[U'(c_1)]$

Can eat today or saving, getting interest, and eating tomorrow

- Eating gives $U'(c_0)$
- Saving gives $(1+r)\delta\mathbb{E}[U'(c_1)]$ (on average)

Must be indifferent for last unit of wealth

- Else would consume something different from (c_0, c_1)

RE-WRITE ($\star$)

Using Euler equation can re-write asset-pricing equation ($\star$)

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Concept check: If $\mathbb{Q}[\omega] > \mathbb{P}[\omega]$ is consumption high or low in state ω?

COVARIANCE FORMULA

Use $\mathrm{Cov}[X, Y] = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]$:

$$X_0 = \frac{1}{1+r}\left(\mathbb{E}[X_1] + \mathrm{Cov}\left[\frac{U_1'}{\mathbb{E}[U_1']}, X_1\right]\right)$$

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Securities valued at expected value plus premium for covariance with U'

NB: Premium for negative covariance with c_1 since U_1' decreasing

Expected excess return formulation for $R_X := \dfrac{X_1}{X_0}$:

$$\mathbb{E}[R_X] - (1+r) = -\text{Cov}\left[\frac{U_1'}{\mathbb{E}[U_1']}, R_X\right]$$

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$$\implies \mathbb{E}[R_X] - (1+r) = \beta_X\big(\mathbb{E}[R_M] - (1+r)\big) \text{ where } \beta_X := \frac{\text{Cov}[R_M, R_X]}{\text{Var}[R_M]}$$

EXAMPLES

EXAMPLE: FAIR COIN

X is 0 if heads or 1 if tails on a fair coin flip

Is $\mathbb{E}[R_X] \leqslant 1 + r$?

EXAMPLE: MARKET

X pays off 1 if market goes up an zero otherwise

Is $\mathbb{E}[R_X] \lesseqgtr 1 + r$?

EXAMPLE: VIRUSES

X pays off 1 if a pandemic is declared in 2023 and zero otherwise

Is $\mathbb{E}[R_X] \lesseqgtr 1 + r$?

EXAMPLE: INTEREST RATES

X pays off 1 if interest rates go up and zero otherwise

Is $\mathbb{E}[R_X] \lesseqgtr 1 + r$?

RECAP

LECTURE 10: RECAP

Utilility maximization

Portfolio choice

Asset pricing (stochastic discount factor)

U' and $\mathbb{Q}$

CAPM

Examples

FIN